

WR503 Wafer 搬運機器人震動評估報告 V2

研究目的

比較2026年7月底機構調整前（3月、6月）與調整後（8月）的上手 / 下手原始震動資料，量化改善或退步幅度。

規格背景

廠商目標為搬運 265 片 wafer 時加速度低於 $\pm 0.3\text{ g}$ ；目前資料皆為 250 片條件。

標準化原則

所有資料先統一至 200 Hz低通頻寬與 512 Hz取樣率；各比較組再使用相同起點與完全相同樣本數。

判讀限制

跨日期能量包絡相關性不足，無法證明錄到相同動作或相同wafer握持區間。因此改善率是經取樣、頻寬及長度標準化後的條件式工程指標，不能取代265片正式驗收。

資料與分析方法

資料範圍

3月：上手、下手各一筆。

6月：WPH249與WPH250，上手、下手各一筆。

8月：濾波前上手J5、濾波前下手J4各一筆。

量化指標

RMS、|g| P99/P99.9、最大絕對值、低於0.3 g樣本比例、超標累積時間、每分鐘超標事件數及最長連續超標時間。

公平比較

全部資料使用四階零相位 200 Hz低通，再重採樣至 512 Hz。每組比較截取相同樣本數；事件率按每分鐘正規化。

改善率定義

對於越低越好的指標：改善率 = (benchmark - August) / benchmark x 100%。正值表示改善，負值表示退步。

資料完整性與處理狀態稽核

missing/duplicate/gap皆為0表示檔案中段無timestamp裁切；起訖流程仍需事件標記確認

dataset	inferred_fs	samples	duration_s	missing_by_timeline	duplicate_timestamps	large_gaps	processing_label
2026-03-26 上手	1024.001	402432	392.999	0	0	0	raw/unfiltered
2026-03-26 下手	512.000	201216	392.998	0	0	0	raw/unfiltered
2026-06-03 WPH249 上手	1023.992	374784	365.999	0	0	0	raw/unfiltered
2026-06-03 WPH249 下手	1023.992	374272	365.499	0	0	0	raw/unfiltered
2026-06-03 WPH250 上手	1023.992	377856	368.999	0	0	0	raw/unfiltered
2026-06-03 WPH250 下手	1024.001	374784	365.999	0	0	0	raw/unfiltered
2026-08-27 上手 J5	1024.001	359936	351.499	0	0	0	raw/unfiltered
2026-08-27 下手 J4	1024.001	363520	354.999	0	0	0	raw/unfiltered
2026-08-27 上手 J5 濾波後	1024.001	359936	351.499	0	0	0	filtered
2026-08-27 下手 J4 濾波後	1024.001	355328	346.999	0	0	0	filtered

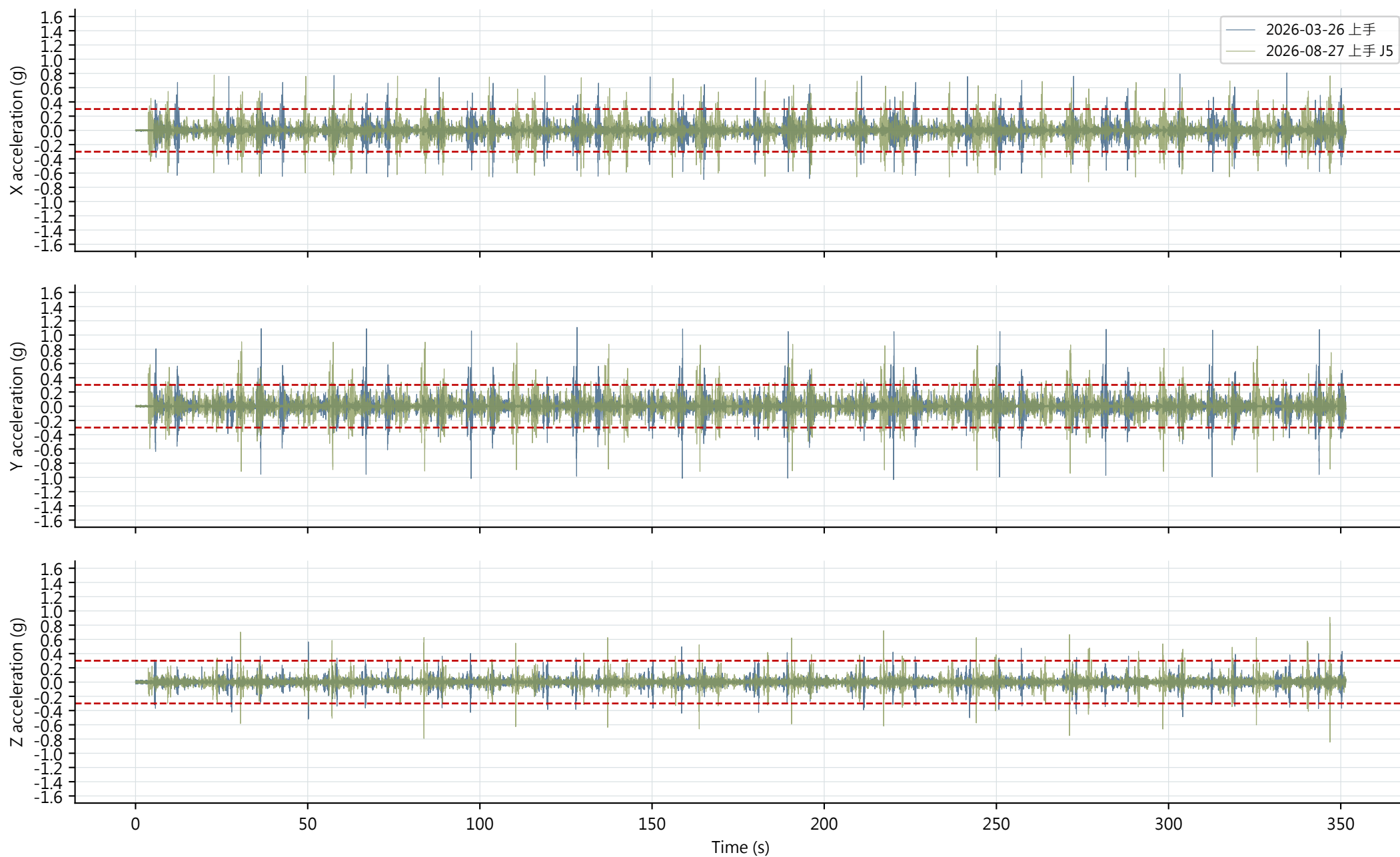
流程能量包絡對齊稽核

相關性 ≥ 0.5 才視為可自動對齊；跨日期比較皆未通過，0827濾波前後則呈現相似流程但非同次錄製

comparison	candidate_lag_s	envelope_correlation	overlap_s	alignment_gate
0326 upper vs 0827 raw upper	-21.50	0.075	330.00	FAIL
0326 lower vs 0827 raw lower	-97.50	0.079	257.50	FAIL
0603 WPH249 upper vs 0827 raw upper	-72.75	0.094	278.75	FAIL
0603 WPH250 upper vs 0827 raw upper	-95.00	0.095	256.50	FAIL
0603 WPH249 lower vs 0827 raw lower	-72.00	0.090	283.00	FAIL
0603 WPH250 lower vs 0827 raw lower	-102.00	0.104	253.00	FAIL
0827 filtered vs raw upper	-79.25	0.816	272.25	PASS
0827 filtered vs raw lower	-79.25	0.797	275.75	PASS

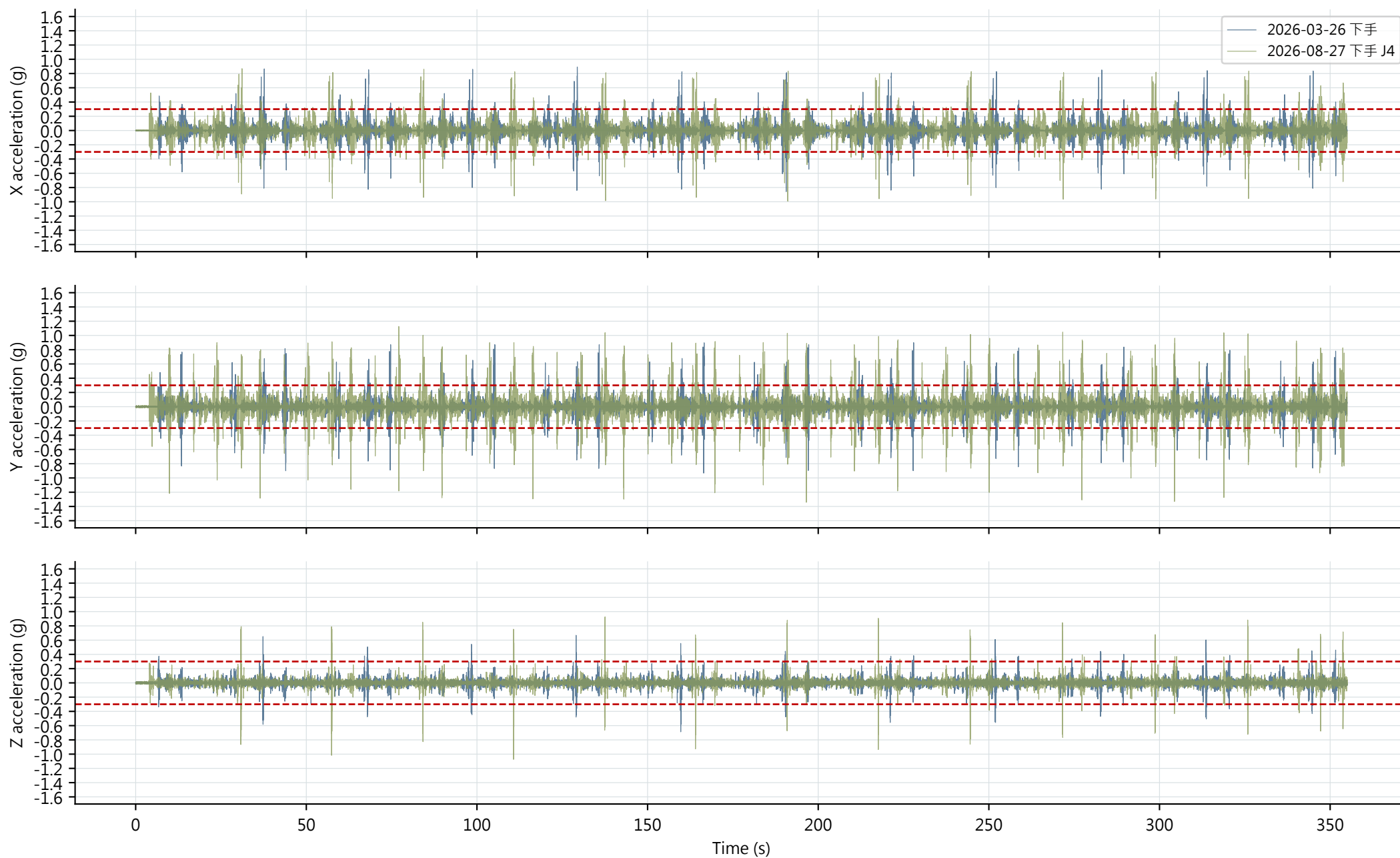
上手：3月基準 vs 8月調整後

原始資料經共同 200 Hz 頻寬、512 Hz重採樣、同長 351.5 s；門檻 ± 0.3 g



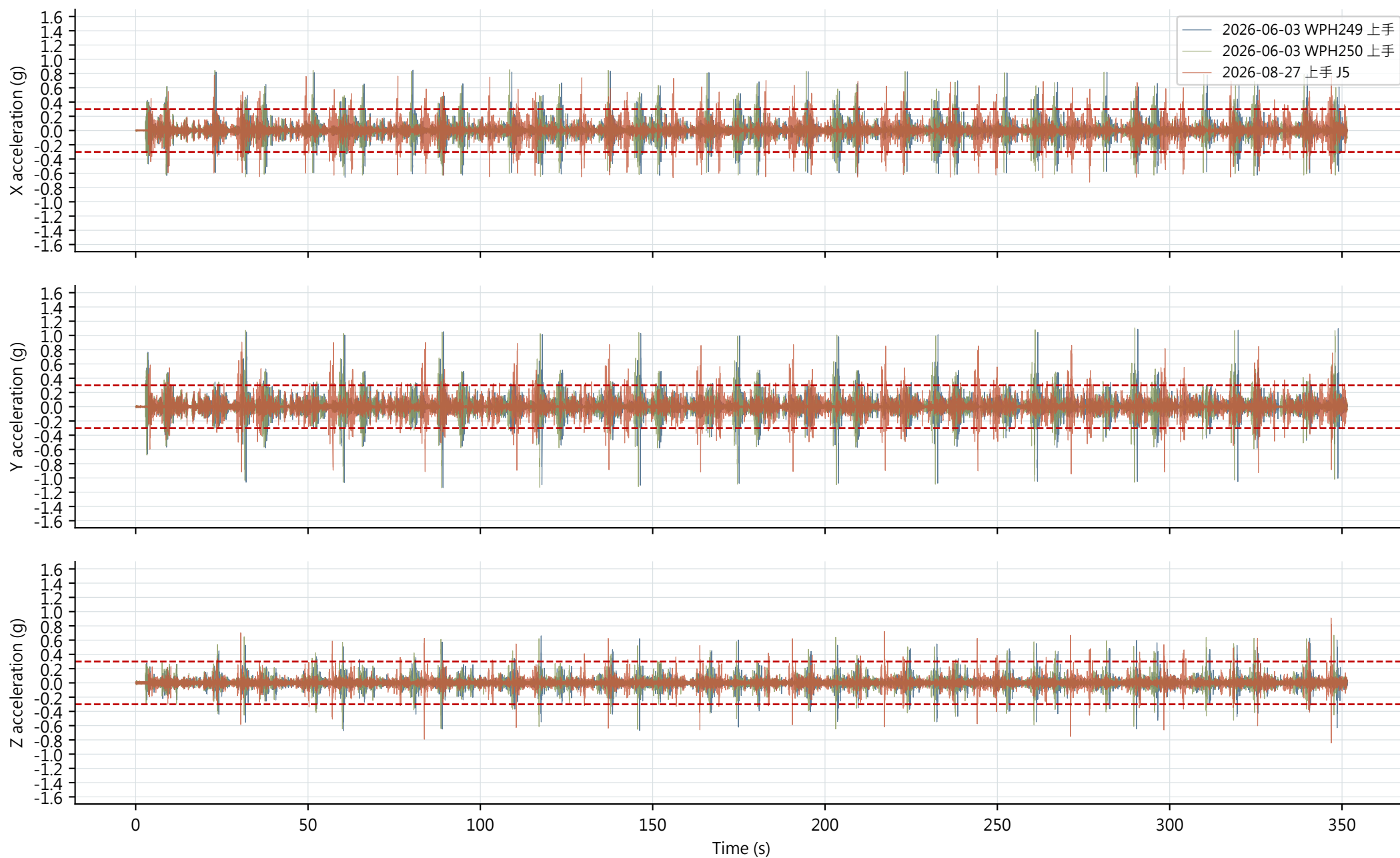
下手：3月基準 vs 8月調整後

原始資料經共同 200 Hz 頻寬、512 Hz重採樣、同長 355.0 s；門檻 ± 0.3 g



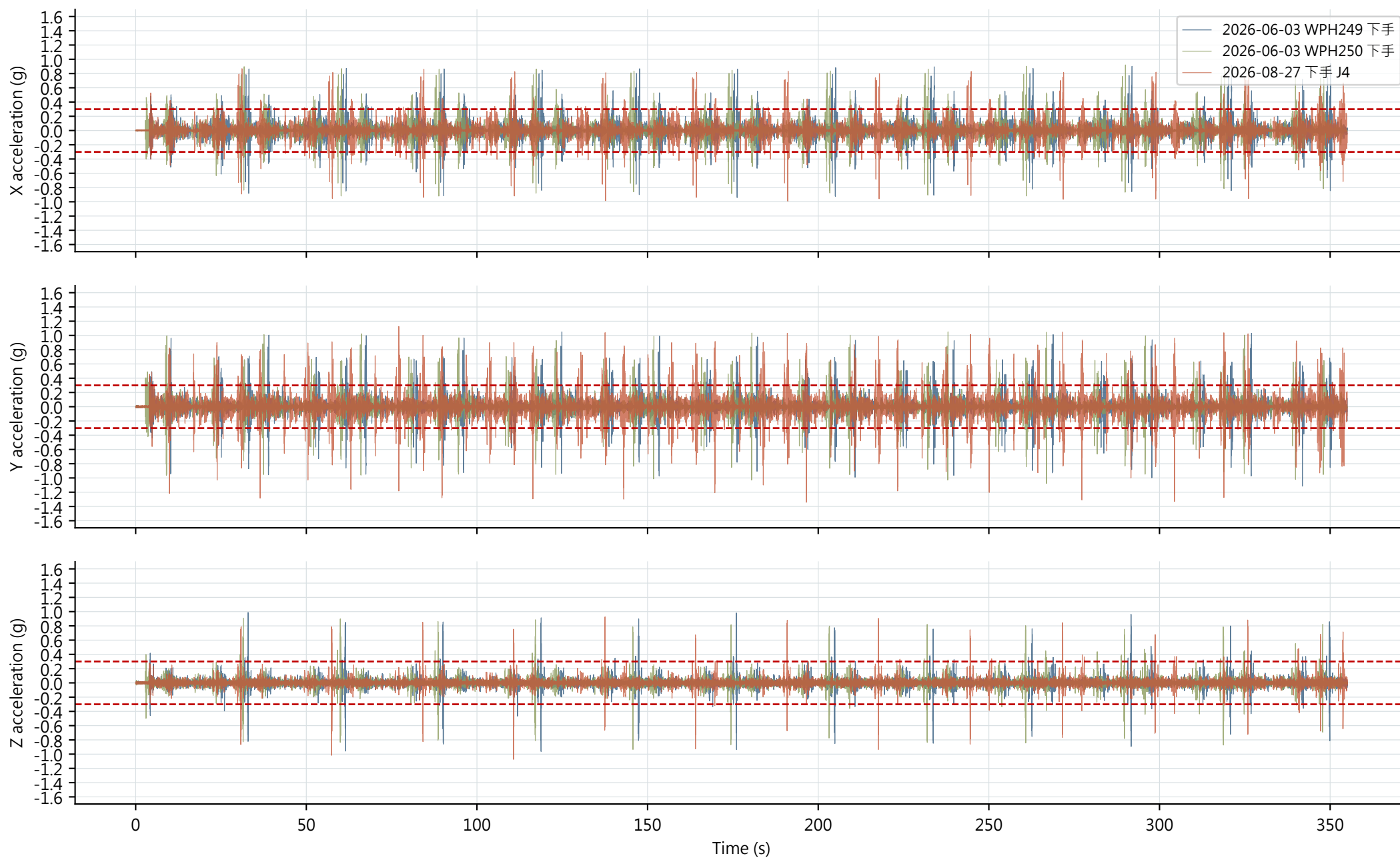
上手：6月基準 vs 8月調整後

原始資料經共同 200 Hz 頻寬、512 Hz重採樣、同長 351.5 s；門檻 ± 0.3 g



下手：6月基準 vs 8月調整後

原始資料經共同 200 Hz 頻寬、512 Hz重採樣、同長 355.0 s；門檻 ± 0.3 g



2026-08-27 (250片) 標準化量化結果

共同 200 Hz頻寬與 512 Hz取樣率；此頁不是265片驗收結果

dataset	axis	RMS_g	P99_abs_g	P99.9_abs_g	Max_abs_g	Within_pct	Exceed_pct	Exceed_seconds	Events_per_min	Max_event_ms
2026-08-27 上手 J5	X	0.099	0.392	0.635	0.795	97.647	2.353	8.270	138.947	74.219
2026-08-27 上手 J5	Y	0.105	0.403	0.829	0.943	97.768	2.232	7.846	99.346	103.516
2026-08-27 上手 J5	Z	0.053	0.218	0.453	1.151	99.627	0.373	1.312	82.959	29.297
2026-08-27 下手 J4	X	0.108	0.413	0.796	0.991	97.559	2.441	8.666	113.915	109.375
2026-08-27 下手 J4	Y	0.136	0.568	0.963	1.585	95.501	4.499	15.971	305.070	44.922
2026-08-27 下手 J4	Z	0.055	0.203	0.669	1.074	99.509	0.491	1.742	73.352	41.016

3月基準 → 8月：上手改善率 — auditable change calculation

Common window: 351.500 s, 179,968 samples at 512 Hz; baseline = single run

Axis	Metric	Baseline raw run(s)	Baseline denominator	2026-08-27 numerator	Formula and result
X	RMS	0.08799 g	0.08799 g	0.09923 g	$(0.08799 \text{ g} - 0.09923 \text{ g}) / 0.08799 \text{ g} = -12.77\%$ (worsened)
X	g P99	0.38082 g	0.38082 g	0.39240 g	$(0.38082 \text{ g} - 0.39240 \text{ g}) / 0.38082 \text{ g} = -3.04\%$ (worsened)
X	g maximum	0.80882 g	0.80882 g	0.79525 g	$(0.80882 \text{ g} - 0.79525 \text{ g}) / 0.80882 \text{ g} = +1.68\%$ (improved)
X	Samples >= 0.3 g	3227/179968 (1.79310%)	1.79310%	4234/179968 (2.35264%)	$(1.79310\% - 2.35264\%) / 1.79310\% = -31.21\%$ (worsened)
X	Exceedance events	413 events (70.498/min)	70.498/min	814 events (138.947/min)	$(70.498/\text{min} - 138.947/\text{min}) / 70.498/\text{min} = -97.09\%$ (worsened)
Y	RMS	0.09104 g	0.09104 g	0.10545 g	$(0.09104 \text{ g} - 0.10545 \text{ g}) / 0.09104 \text{ g} = -15.82\%$ (worsened)
Y	g P99	0.37494 g	0.37494 g	0.40293 g	$(0.37494 \text{ g} - 0.40293 \text{ g}) / 0.37494 \text{ g} = -7.47\%$ (worsened)
Y	g maximum	1.12421 g	1.12421 g	0.94316 g	$(1.12421 \text{ g} - 0.94316 \text{ g}) / 1.12421 \text{ g} = +16.10\%$ (improved)
Y	Samples >= 0.3 g	2967/179968 (1.64863%)	1.64863%	4017/179968 (2.23206%)	$(1.64863\% - 2.23206\%) / 1.64863\% = -35.39\%$ (worsened)
Y	Exceedance events	541 events (92.347/min)	92.347/min	582 events (99.346/min)	$(92.347/\text{min} - 99.346/\text{min}) / 92.347/\text{min} = -7.58\%$ (worsened)
Z	RMS	0.04295 g	0.04295 g	0.05277 g	$(0.04295 \text{ g} - 0.05277 \text{ g}) / 0.04295 \text{ g} = -22.85\%$ (worsened)
Z	g P99	0.18788 g	0.18788 g	0.21813 g	$(0.18788 \text{ g} - 0.21813 \text{ g}) / 0.18788 \text{ g} = -16.10\%$ (worsened)
Z	g maximum	0.59499 g	0.59499 g	1.15079 g	$(0.59499 \text{ g} - 1.15079 \text{ g}) / 0.59499 \text{ g} = -93.41\%$ (worsened)
Z	Samples >= 0.3 g	337/179968 (0.18726%)	0.18726%	672/179968 (0.37340%)	$(0.18726\% - 0.37340\%) / 0.18726\% = -99.41\%$ (worsened)
Z	Exceedance events	313 events (53.428/min)	53.428/min	486 events (82.959/min)	$(53.428/\text{min} - 82.959/\text{min}) / 53.428/\text{min} = -55.27\%$ (worsened)

3月基準 → 8月：下手改善率 — auditable change calculation

Common window: 355.000 s, 181,760 samples at 512 Hz; baseline = single run

Axis	Metric	Baseline raw run(s)	Baseline denominator	2026-08-27 numerator	Formula and result
X	RMS	0.09409 g	0.09409 g	0.10803 g	$(0.09409 \text{ g} - 0.10803 \text{ g}) / 0.09409 \text{ g} = -14.81\%$ (worsened)
X	g P99	0.36183 g	0.36183 g	0.41349 g	$(0.36183 \text{ g} - 0.41349 \text{ g}) / 0.36183 \text{ g} = -14.28\%$ (worsened)
X	g maximum	0.89227 g	0.89227 g	0.99093 g	$(0.89227 \text{ g} - 0.99093 \text{ g}) / 0.89227 \text{ g} = -11.06\%$ (worsened)
X	Samples >= 0.3 g	2870/181760 (1.57901%)	1.57901%	4437/181760 (2.44113%)	$(1.57901\% - 2.44113\%) / 1.57901\% = -54.60\%$ (worsened)
X	Exceedance events	450 events (76.056/min)	76.056/min	674 events (113.915/min)	$(76.056/\text{min} - 113.915/\text{min}) / 76.056/\text{min} = -49.78\%$ (worsened)
Y	RMS	0.09285 g	0.09285 g	0.13590 g	$(0.09285 \text{ g} - 0.13590 \text{ g}) / 0.09285 \text{ g} = -46.35\%$ (worsened)
Y	g P99	0.40556 g	0.40556 g	0.56772 g	$(0.40556 \text{ g} - 0.56772 \text{ g}) / 0.40556 \text{ g} = -39.98\%$ (worsened)
Y	g maximum	0.93234 g	0.93234 g	1.58504 g	$(0.93234 \text{ g} - 1.58504 \text{ g}) / 0.93234 \text{ g} = -70.01\%$ (worsened)
Y	Samples >= 0.3 g	3711/181760 (2.04170%)	2.04170%	8177/181760 (4.49879%)	$(2.04170\% - 4.49879\%) / 2.04170\% = -120.34\%$ (worsened)
Y	Exceedance events	667 events (112.732/min)	112.732/min	1805 events (305.070/min)	$(112.732/\text{min} - 305.070/\text{min}) / 112.732/\text{min} = -170.61\%$ (worsened)
Z	RMS	0.04641 g	0.04641 g	0.05507 g	$(0.04641 \text{ g} - 0.05507 \text{ g}) / 0.04641 \text{ g} = -18.65\%$ (worsened)
Z	g P99	0.19158 g	0.19158 g	0.20343 g	$(0.19158 \text{ g} - 0.20343 \text{ g}) / 0.19158 \text{ g} = -6.19\%$ (worsened)
Z	g maximum	0.68741 g	0.68741 g	1.07405 g	$(0.68741 \text{ g} - 1.07405 \text{ g}) / 0.68741 \text{ g} = -56.25\%$ (worsened)
Z	Samples >= 0.3 g	609/181760 (0.33506%)	0.33506%	892/181760 (0.49076%)	$(0.33506\% - 0.49076\%) / 0.33506\% = -46.47\%$ (worsened)
Z	Exceedance events	516 events (87.211/min)	87.211/min	434 events (73.352/min)	$(87.211/\text{min} - 73.352/\text{min}) / 87.211/\text{min} = +15.89\%$ (improved)

6月平均基準 → 8月：上手改善率 — auditable change calculation

Common window: 351.500 s, 179,968 samples at 512 Hz; baseline = mean of WPH249 and WPH250; both raw values shown

Axis	Metric	Baseline raw run(s)	Baseline denominator	2026-08-27 numerator	Formula and result
X	RMS	0.09799 g / 0.09791 g	0.09795 g	0.09923 g	(0.09795 g - 0.09923 g) / 0.09795 g = -1.31% (worsened)
X	g P99	0.43116 g / 0.43034 g	0.43075 g	0.39240 g	(0.43075 g - 0.39240 g) / 0.43075 g = +8.90% (improved)
X	g maximum	0.85678 g / 0.85423 g	0.85551 g	0.79525 g	(0.85551 g - 0.79525 g) / 0.85551 g = +7.04% (improved)
X	Samples >= 0.3 g	4547/179968 (2.52656%) / 4566/179968 (2.53712%)	2.53184%	4234/179968 (2.35264%)	(2.53184% - 2.35264%) / 2.53184% = +7.08% (improved)
X	Exceedance events	698 events (119.147/min) / 689 events (117.610/min)	118.378/min	814 events (138.947/min)	(118.378/min - 138.947/min) / 118.378/min = -17.38% (worsened)
Y	RMS	0.10115 g / 0.10134 g	0.10124 g	0.10545 g	(0.10124 g - 0.10545 g) / 0.10124 g = -4.15% (worsened)
Y	g P99	0.40867 g / 0.41099 g	0.40983 g	0.40293 g	(0.40983 g - 0.40293 g) / 0.40983 g = +1.68% (improved)
Y	g maximum	1.13591 g / 1.13825 g	1.13708 g	0.94316 g	(1.13708 g - 0.94316 g) / 1.13708 g = +17.05% (improved)
Y	Samples >= 0.3 g	3709/179968 (2.06092%) / 3754/179968 (2.08593%)	2.07342%	4017/179968 (2.23206%)	(2.07342% - 2.23206%) / 2.07342% = -7.65% (worsened)
Y	Exceedance events	666 events (113.684/min) / 657 events (112.148/min)	112.916/min	582 events (99.346/min)	(112.916/min - 99.346/min) / 112.916/min = +12.02% (improved)
Z	RMS	0.05308 g / 0.05289 g	0.05299 g	0.05277 g	(0.05299 g - 0.05277 g) / 0.05299 g = +0.41% (improved)
Z	g P99	0.22048 g / 0.22179 g	0.22114 g	0.21813 g	(0.22114 g - 0.21813 g) / 0.22114 g = +1.36% (improved)
Z	g maximum	0.70911 g / 0.69106 g	0.70008 g	1.15079 g	(0.70008 g - 1.15079 g) / 0.70008 g = -64.38% (worsened)
Z	Samples >= 0.3 g	811/179968 (0.45064%) / 806/179968 (0.44786%)	0.44925%	672/179968 (0.37340%)	(0.44925% - 0.37340%) / 0.44925% = +16.88% (improved)
Z	Exceedance events	652 events (111.294/min) / 652 events (111.294/min)	111.294/min	486 events (82.959/min)	(111.294/min - 82.959/min) / 111.294/min = +25.46% (improved)

6月平均基準 → 8月：下手改善率 — auditable change calculation

Common window: 355.000 s, 181,760 samples at 512 Hz; baseline = mean of WPH249 and WPH250; both raw values shown

Axis	Metric	Baseline raw run(s)	Baseline denominator	2026-08-27 numerator	Formula and result
X	RMS	0.10344 g / 0.10351 g	0.10347 g	0.10803 g	(0.10347 g - 0.10803 g) / 0.10347 g = -4.40% (worsened)
X	g P99	0.41189 g / 0.41259 g	0.41224 g	0.41349 g	(0.41224 g - 0.41349 g) / 0.41224 g = -0.30% (worsened)
X	g maximum	0.94141 g / 0.95338 g	0.94740 g	0.99093 g	(0.94740 g - 0.99093 g) / 0.94740 g = -4.60% (worsened)
X	Samples >= 0.3 g	4171/181760 (2.29478%) / 4185/181760 (2.30249%)	2.29864%	4437/181760 (2.44113%)	(2.29864% - 2.44113%) / 2.29864% = -6.20% (worsened)
X	Exceedance events	771 events (130.310/min) / 778 events (131.493/min)	130.901/min	674 events (113.915/min)	(130.901/min - 113.915/min) / 130.901/min = +12.98% (improved)
Y	RMS	0.10659 g / 0.10699 g	0.10679 g	0.13590 g	(0.10679 g - 0.13590 g) / 0.10679 g = -27.26% (worsened)
Y	g P99	0.47531 g / 0.47424 g	0.47478 g	0.56772 g	(0.47478 g - 0.56772 g) / 0.47478 g = -19.57% (worsened)
Y	g maximum	1.11533 g / 1.11149 g	1.11341 g	1.58504 g	(1.11341 g - 1.58504 g) / 1.11341 g = -42.36% (worsened)
Y	Samples >= 0.3 g	5296/181760 (2.91373%) / 5260/181760 (2.89393%)	2.90383%	8177/181760 (4.49879%)	(2.90383% - 4.49879%) / 2.90383% = -54.93% (worsened)
Y	Exceedance events	859 events (145.183/min) / 830 events (140.282/min)	142.732/min	1805 events (305.070/min)	(142.732/min - 305.070/min) / 142.732/min = -113.74% (worsened)
Z	RMS	0.05808 g / 0.05669 g	0.05739 g	0.05507 g	(0.05739 g - 0.05507 g) / 0.05739 g = +4.04% (improved)
Z	g P99	0.21161 g / 0.20838 g	0.21000 g	0.20343 g	(0.21000 g - 0.20343 g) / 0.21000 g = +3.13% (improved)
Z	g maximum	1.08867 g / 1.01485 g	1.05176 g	1.07405 g	(1.05176 g - 1.07405 g) / 1.05176 g = -2.12% (worsened)
Z	Samples >= 0.3 g	998/181760 (0.54908%) / 927/181760 (0.51001%)	0.52954%	892/181760 (0.49076%)	(0.52954% - 0.49076%) / 0.52954% = +7.32% (improved)
Z	Exceedance events	486 events (82.141/min) / 537 events (90.761/min)	86.451/min	434 events (73.352/min)	(86.451/min - 73.352/min) / 86.451/min = +15.15% (improved)

V4 calculation traceability summary

Authoritative result pages

The four preceding comparison tables show every X/Y/Z result as baseline raw value(s), baseline denominator, August numerator, and $(\text{baseline} - \text{August}) / \text{baseline}$. For June, WPH249 and WPH250 are displayed separately before their mean is used.

Event and threshold counts

Exceedance events are reported as true counts together with the common observation duration and events/min. Threshold sample rates show exceeded samples / total samples. Positive values mean improvement; negative values mean worsening.

整體判斷

在共同頻寬、取樣率與紀錄長度下，7月底機構調整仍未呈現所有手臂與軸向的一致改善；下手Y仍是優先調查項目。

規格判定

8月250片資料的六個軸向均出現 $\geq 0.3\text{ g}$ 樣本，因此若規格定義為任何瞬時樣本皆須低於0.3 g，現況不符合；且尚未完成265片負載驗證。

下一步

1. 與廠商凍結0.3 g的計算定義與濾波頻帶。
2. 在相同軌跡、速度、sensor orientation與取樣率下重測250片。
3. 優先定位下手Y的振源與超標事件發生動作段。
4. 改善確認後，以265片進行至少三次重複驗收。